# FinInteract: A Bilingual Benchmark for Evaluating Financial Search Agents on Ambiguous Queries

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## ABSTRACT

Financial question answering benchmarks assume that every query is fully specified, yet real analyst queries routinely carry unresolved ambiguity: the same question—*"What was Meta Platforms' operating income?"*—can yield answers differing by 34% depending on whether the scope is consolidated total or the Family of Apps segment. We introduce FinInteract, a bilingual (English/Chinese) benchmark of 500 instances that pairs each question $Q$ with a *default interpretation* (the answer a non-expert would assume) and an *intended interpretation* (uniquely identified by a disambiguating context $C$). Instances span five axes of financial ambiguity—temporal scope, metric definition, entity scope, filing vintage, and recognition policy—drawn from SEC EDGAR filings and A-share annual reports. We evaluate eight frontier models under three ReAct-based modes (Answer-only, Answer+Search, Answer+Search+Interact) using the Disambiguation Efficiency metric $\mathrm{DisE}^{+}_{\mathrm{all}}$, which rewards efficient clarification. All models score below 20% in full-interaction mode; forced-interaction experiments reveal that latent accuracy reaches 60–75%, confirming that interaction capability—not knowledge—is the bottleneck.

## CCS CONCEPTS

• **Information systems → Question answering**; • **Computing methodologies →** *Natural language processing*.

## KEYWORDS

financial question answering, ambiguity resolution, interactive agents, benchmark, disambiguation, large language models

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## 1 INTRODUCTION

Large language models are increasingly deployed in high-stakes financial workflows: earnings-call analysis, SEC filing review, analyst report generation, and investor query handling. Yet the benchmarks used to evaluate these models share a critical blind spot—they assume that every query is fully specified before the model begins to answer.

Consider the question *"What was Meta Platforms' operating income?"* This question is genuinely ambiguous: should *operating income* be the consolidated GAAP figure for the total company ($46.75B for FY2023), or the Family of Apps segment figure ($62.87B)? Both are correct under their respective interpretations; the answers differ by 34%. A skilled financial analyst would not guess—they would ask a clarifying question. Existing benchmarks reward the model that guesses correctly; they do not distinguish a model that recognizes the ambiguity from one that happens to pick the default interpretation.

**Table 1: Running example: a FinInteract instance. The same question yields two defensible answers. Without the disambiguating context $C$, a model's default interpretation leads to the wrong answer.**

| Field | Value |
|---|---|
| $Q$ | What was Meta Platforms' operating income? |
| $C$ | Total company (consolidated) results under U.S. GAAP for the year ended December 31, 2023. |
| $A$ (intended) | $46.75B *[consolidated GAAP]* |
| $A_d$ (default) | $62.87B *[Family of Apps segment]* |
| Ambiguity axis | Entity scope |
| $H_0$ | 1.58 bits |
| Intended span | "Income from operations for 2023 was $46.75 billion..." |
| Default span | "Family of Apps...Income from operations $62,871..." |

*Limitations of Existing Benchmarks.* FinanceBench [3] provides 150 expert-curated questions over SEC filings, each with a unique correct answer verified against the source document; queries are pre-disambiguated by construction. FinQA [2] and DocFinQA [8] target numerical reasoning over earnings reports but give the model the exact evidence table alongside the question, bypassing retrieval and interaction entirely. BizBench [4] and FinBen [11] aggregate multiple financial tasks but treat QA as closed-book or retrieval-augmented reading comprehension. None of these benchmarks ask whether a model *recognizes* that a question is ambiguous, nor whether it can *resolve* that ambiguity through strategic interaction with a user.

*The Ambiguous-QA Gap in Finance.* The ambiguous-QA literature has matured in the general domain: AmbigQA [7] introduced multi-answer formulations; CondAmbigQA [6] added condition-structured annotations; CLAM [5] demonstrated that clarification


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pipelines improve accuracy on ambiguous queries. However, all of this work operates on general-domain Wikipedia questions. Finance is structurally different in three ways. First, financial ambiguity is *endemic and systematic*—a single number such as "net income" can legitimately differ by 30–40% depending on reporting scope, accounting basis, and filing vintage. Second, *misanswers are costly*—an investor acting on the wrong revenue figure or a compliance officer citing the wrong filing version can cause real financial harm. Third, *ground truth is programmatically verifiable*—SEC filings and XBRL facts provide exact machine-readable values for every metric, period, and scope.

*The Interaction Stagnation Hypothesis.* InteractComp [1] established an important longitudinal finding: while retrieval accuracy on web search benchmarks has improved rapidly with model scale, interaction capability has not kept pace. Across eight frontier models on general-domain tasks, fewer than 14% of questions were answered correctly in full-interaction mode, even though forced-interaction experiments showed the same models could achieve 60–70% accuracy when required to elicit disambiguating context. We hypothesize that this stagnation is more severe in finance, where *domain knowledge is required to even recognize the axes of ambiguity*: a model that does not know what "non-GAAP" means cannot recognize that "adjusted net income" is under-specified.

*Our Proposal: FinInteract.* We adapt the InteractComp methodology to the finance domain, introduce a five-axis financial ambiguity taxonomy, and release FinInteract —a bilingual benchmark of 500 instances (400 English / 100 Chinese) designed to measure whether financial search agents can recognize query ambiguity and resolve it through efficient interaction.

**Contributions:**

(1) **Benchmark**: 500 bilingual instances with paired default/intended interpretations, both grounded in verbatim filing quotes, verified by adversarial LLM trials and human annotation (§4).
(2) **Error Taxonomy**: A seven-type error decomposition framework (E1–E7) that maps observed evaluation signals to actionable failure modes (ambiguity blindness, wrong-axis clarification, retrieval dominance, grounding failure) and links them to existing metrics (§7).
(3) **Method Baseline**: Axis-Aware Clarification ReAct —a three-component intervention (ambiguity pre-check, axis-conditioned clarification, interpretation-state-gated answering) that directly targets E1–E3 errors and is evaluated against standard ReAct and oracle baselines (§7).

# 2 TASK DEFINITION

## 2.1 Problem Formulation

Each FinInteract instance is a tuple ($Q$, $C$, $A$, $A_d$, $\mathcal{I}_{\text{int}}$, $\mathcal{I}_{\text{def}}$):

- $Q$ is an ambiguous financial question over a public company filing.
- $C$ is a disambiguating context: the minimal set of scope or definitional constraints that uniquely collapses $Q$ to $A$.
- $A$ is the correct answer under the *intended* interpretation $\mathcal{I}_{\text{int}}$.
- $A_d$ is the correct answer under the *default* interpretation $\mathcal{I}_{\text{def}}$ (the answer a non-expert reading $Q$ alone would assume), with $A_d \neq A$.
- $\mathcal{I}_{\text{int}}$, $\mathcal{I}_{\text{def}}$ are structured records specifying entity scope, period, metric, and reporting basis.

An agent reading $Q$ alone (without $C$) must find it genuinely ambiguous—both $A$ and $A_d$ are defensible from publicly available filings. This *default-vs-intended pairing* generalizes InteractComp's target-distractor design to the financial domain.

## 2.2 Design Principles

**Easy to verify, Ambiguous to resolve.** Adopted from InteractComp: ground truth is programmatically verifiable against SEC XBRL facts and CNINFO-structured data, yet the question cannot be correctly resolved without the disambiguating context $C$.

**No yes/no questions.** $Q$ must elicit a substantive financial value, not a binary confirmation.

**Company must be named in $Q$.** The question is anchored to a specific issuer (English: capitalized proper noun; Chinese: CJK company name).

**Context must not contain the answer.** $C$ specifies scope constraints but not the numeric value $A$ itself.

## 2.3 Agent Interaction Model

We follow InteractComp's ReAct framework [12]. The agent may issue three types of actions: **search**($q$)—retrieves a passage from the filing corpus; **interact**($q$)—presents a yes/no question to a simulated user who answers based on $C$; **answer**($v$)—submits a final answer. We evaluate agents in three primary modes (Table 2) and four ablation modes.

**Table 2: Evaluation modes.**

| Mode | Actions | Description |
|---|---|---|
| Answer-only | answer | Pure parametric recall |
| Answer+Search | search, answer | Oracle retrieval augmented |
| Answer+Search+Interact | search, interact, answer | Standard full interaction |
| **Axis-Aware ReAct** | state, search, interact, answer | **Proposed: axis-conditione** |
| Always-ask | Must ask ≥ 1 question before answering | |
| Axis-oracle | Told the primary axis, not the value | |
| Template-oracle | Fixed human-written question per axis | |
| Enumerate | Lists all interpretations and answers | |

# 3 FINANCIAL AMBIGUITY TAXONOMY

We define five axes of financial ambiguity, each grounded in standard accounting and disclosure practice (Table 3).

Each instance exercises one primary axis (mandatory) and optionally one or two secondary axes where the passage genuinely supports additional ambiguity. The target distribution over primary axes is 25% temporal scope, 25% metric definition, 20% recognition policy, 15% entity scope, and 15% filing vintage—calibrated to counterbalance the over-representation of entity scope (66% of pool passages) and filing vintage (67%) in the raw passage pool.

*Disambiguation Entropy.* For each instance we compute the prior disambiguation entropy: $H_0 = \log_2 k$, where $k$ is the number of plausible interpretations before any clarifying interaction. For a single-axis instance with two interpretations, $H_0 = 1$ bit. For a

**Table 3: Five-axis financial ambiguity taxonomy.**

| Axis | Definition |
|---|---|
| **Temporal scope** | Ambiguity over which time period is intended: fiscal year ending September vs. calendar year; Q3 results vs. full year; TTM vs. point-in-time. |
| **Metric definition** | Ambiguity over accounting basis: GAAP net income vs. non-GAAP adjusted income; organic revenue vs. as-reported; operating EBITDA vs. net income. |
| **Entity scope** | Ambiguity over consolidation level: segment vs. consolidated total; parent-attributable vs. full consolidated; Class A vs. Class B shares; A-shares vs. H-shares. |
| **Filing vintage** | Ambiguity over filing version: original 10-K vs. amended 10-K/A; as-reported vs. restated; preliminary earnings release vs. final filing. |
| **Recognition policy** | Ambiguity over accounting policy: revenue recognized point-in-time vs. over time; gross vs. net revenue; capitalized vs. expensed R&D. |

two-axis instance with three and four plausible values respectively, $H_0 = \log_2(3\times4) \approx 3.58$ bits. This quantity anchors our $\text{DisE}^+_{\text{all}}$ metric (§5.3).

# 4 DATASET CONSTRUCTION

## 4.1 Data Sources

**English (EN, ~80% of pool).** *DocFinQA*—780 long-context passages extracted from 10-K and 10-Q filings, with verified candidate answers from annotated QA pairs. *EDGAR (XBRL-derived)*—111 passages built from machine-readable XBRL facts for S&P 500 and Russell 1000 companies (FY2024–2025). Because instances are newly constructed from structured facts requiring fresh ambiguity pairs, memorization risk is substantially reduced relative to benchmarks that reuse published QA pairs directly—though we do not claim absolute contamination safety.

**Chinese (ZH, ~20% of pool).** *CNINFO/akshare*—237 passages derived from A-share annual reports (上证 50 + 沪深 300 constituents) via structured financial statement APIs. Three passage types: (1) metric definition—扣非净利润 vs. 归母净利润 (non-recurring-excluded profit vs. parent-attributable profit); (2) entity scope—归母净利润 vs. 净利润 (parent-attributable vs. fully consolidated including minority interest); (3) temporal scope—year-over-year revenue growth. ZH instances are expected to be harder for current models due to sparser training data on A-share filings and more complex corporate structure nomenclature (e.g., A/H-share distinctions, 集团 vs. 子公司 scope).

## 4.2 Three-Role Construction Pipeline

Instance construction follows a three-role pipeline. Each passage attempt costs approximately 10–14 minutes of wall time, dominated by two sequential GPT-5 calls.

*Step 1: Constructor (GPT-5, ~2–5 min).* A single GPT-5 call with a 4,096-token budget generates the full $(Q, C, A)$ triple given a filing passage, verified candidate answer, and mandatory primary ambiguity axis. The output JSON includes `question`, `context`, `answer`, `default_answer`, `intended_evidence_span`, `default_evidence_span`, `intended_interpretation`, `default_interpretation`, and `axes_exercised`. Axis-specific instructions are injected per primary axis to prevent the constructor from drifting toward whichever axis is easiest to generate.

*Step 2: Pre-verifier sanity checks (~instant).* Thirteen automated code-level rules are applied without any API calls (Table 4). Any failure immediately rejects the passage, avoiding wasted verifier budget.

*Step 3: Adversarial Verifier (10 trials, parallelized, ~5–10 min).* Ten verifier calls fire simultaneously via a thread pool: five GPT-5-mini trials and five GPT-5 trials. Each trial asks the model to answer $Q$ *without* context $C$ and to state the interpretation it assumed (period, entity, metric, basis). An instance is **rejected** if $\geq 2$ of the 10 trials both (a) produce the correct answer $A$ and (b) state an assumed interpretation that aligns with the intended interpretation. This two-condition criterion prevents false rejections from lucky guesses: if a model produces the right number while assuming a different interpretation (e.g., the default), the coincidence is not counted against the instance. Because all ten calls run concurrently, wall time equals the latency of the slowest single GPT-5 call.

**Table 4: Pre-verifier sanity checks (13 rules).**

| Rule | Description |
|---|---|
| R1 | $Q$ must not be answerable yes/no |
| R2 | $Q$ must not contain disambiguating terms or inline dates |
| R4 | $Q$ must name the company |
| R5 | $Q$ must ask about a substantive financial metric |
| R6 | Answer type consistent with question type |
| R7 | Context $C$ must not contain the answer value |
| R8 | Intended and default interpretations must differ |
| R9 | All axis labels must be valid vocabulary items |
| R10 | Answer must be non-trivial (not empty or "0") |
| R11 | `default_answer` present and differs from `answer` |
| R12 | Both evidence spans non-empty |
| R13 | Evidence spans must be meaningfully distinct (≤85% overlap) |

## 4.3 Axis Diversity Enforcement

A naive construction loop over-represents axes common in the passage pool. We enforce target axis shares through four mechanisms: (1) a priority function with a 3× penalty on over-quota axes; (2) hard exclusion at 2× quota; (3) rarity-adjusted initial sort (passages sorted by $\sum_{\text{axes}}(\text{target\_share}/\text{pool\_frequency})$); and (4) dynamic re-sort every 10 instances.

## 4.4 Human Validation Protocol

Each accepted instance is reviewed by two annotators using an eight-question protocol (H1–H8): H1 *Is $Q$ ambiguous without $C$?* H2 *Is the default interpretation plausible for a non-expert?* H3 *Does $C$ uniquely identify the intended interpretation?* H4 *Does $C$ avoid* *directly stating the answer?* H5 *Is A correct under the intended interpretation?* H6 *Is $A_d$ correct under the default interpretation?* H7 *Is the primary axis label correct?* H8 *What yes/no question would a financial analyst naturally ask to resolve the ambiguity?* (free text, used to compute human AxisHit). Disagreements are adjudicated by the primary author.

# 5 EVALUATION FRAMEWORK

## 5.1 Agent Architecture

We evaluate agents under the ReAct framework in seven configurations (Table 2). The **interact** action presents a yes/no question to a simulated user. The user simulator (GPT-5 at temperature 1.0) is given the disambiguating context $C$ and responds yes, no, or "I don't know." To test robustness, we evaluate under three simulator variants: **LLM** (primary; GPT-5, $T = 1.0$), **Oracle** (deterministic keyword matching against $\mathcal{I}_{\text{int}}$), and **Noisy** (15% random answer corruption).

## 5.2 Models

Eight models are evaluated across all three primary modes: GPT-5, GPT-5-mini, GPT-4o (OpenAI); Claude-Sonnet-4, Claude-Opus-4 (Anthropic); DeepSeek-V3.1, Qwen3-235B, GLM-4.5 (open-weight).

## 5.3 Metrics

**Accuracy (%)** is the fraction of instances answered correctly, graded by GPT-4o-mini with financial-domain tolerance (±1% numeric, entity/ticker equivalence, currency normalization; fiscal-year mismatch = wrong).

**$\text{DisE}^{+}_{\text{all}}$** (primary metric) is:

$$\text{DisE}^{+}_{\text{all}} = \mathbf{1}[\text{correct}] \times \frac{H_0}{\max(n_{\text{asks}}, 1)}$$

Zero when incorrect; rewards confident correct zero-ask answers ($n_{\text{asks}} = 0$ gives $H_0$) and efficient clarification.

**$\text{DisE}^{+}_{\text{interact}}$** (secondary) is:

$$\text{DisE}^{+}_{\text{interact}} = \mathbf{1}[\text{correct}] \times \mathbf{1}[n_{\text{asks}}>0] \times \frac{H_0}{n_{\text{asks}}}$$

Only defined when the agent asked $\geq 1$ question; specifically rewards successful clarification.

**AxisHit (%)**—for each interact action, GPT-4o-mini classifies whether the clarifying question targets a true ambiguity axis for that instance. We report AxisHit@1, AnyAxisHit, WrongAxisRate, and GenericAskRate.

**Additional metrics**: Round (mean turns), IR (interaction rate), and 5-bin Expected Calibration Error.

# 6 EXPERIMENTS

## 6.1 Findings

We report results for GPT-5 and GPT-4o across all three modes (Table 5); the remaining models are in progress. Five findings emerge.

**Finding 1: Interaction is a capability, not a free lever — it helps the strong model and *harms* the weak one.** Adding interaction raises GPT-5 from 6.4% to 20.2% (+13.8), but *lowers* GPT-4o from 12.1% to 4.6% (−7.5). Forcing the weaker model to interact makes it worse than not interacting at all. This extends InteractComp's "interaction stagnation" into a sharper claim: for models that cannot use it, interaction is actively harmful.

**Finding 2: Model rankings flip with the interaction axis.** GPT-4o *outperforms* GPT-5 under pure retrieval (12.1 vs. 6.4) yet is *dominated* by it under interaction (4.6 vs. 20.2). A single-mode benchmark would mis-rank these models; the interaction axis is what separates them, which is precisely the dimension FinInteract is designed to measure.

**Finding 3: Retrieval cannot substitute for interaction.** Oracle retrieval lifts GPT-5 to only 6.4%: because each disambiguating passage contains *both* the intended and default values, retrieval surfaces the ambiguity but does not resolve it. Only a targeted clarification collapses the question to a unique answer, which interaction (20.2%) begins to do.

**Finding 4: The weak model's failure is clarification non-integration, not inability to ask.** GPT-4o asks readily (IR 99%, 4.7 questions on average) but fails to *integrate* the user's responses: 87% of its interactive errors are non-committal refusals (e.g. "please specify the exact period") issued *after* the simulator has answered. It can recognize ambiguity and ask, but cannot close the loop and commit — a multi-turn integration gap, not a protocol artifact (only 2% produced no answer at all).

**Finding 5: Models are severely overconfident on ambiguous queries.** Expected Calibration Error ranges from 0.53 (GPT-5, interact) to 0.95 (GPT-4o, answer-only); models report 85–95% confidence while answering 0–20% correctly. Both models also ask imprecisely — AxisHit@1 is only 0.38 (GPT-5) and 0.37 (GPT-4o), so even when a model asks, it targets the correct ambiguity axis barely a third of the time, motivating the axis-conditioned policy of §7.

*Difficulty and headroom.* Per-instance accuracy falls monotonically with disambiguation entropy $H_0$ (from 10% at $H_0$=1 to under 1% at $H_0 \approx 3.6$), validating $H_0$ as a difficulty proxy; 20 of 173 instances are solved by no model in any mode. The best configuration reaches 20.2%, leaving the benchmark far from saturated.

## 6.2 Ablation: Forced-Interaction Analysis

Following InteractComp, we run a forced-interaction ablation requiring the agent to ask $n \in \{2, 4, 6, 8, 10\}$ clarifying questions before answering. This reveals the *latent capacity* of each model and distinguishes overconfidence (high skip rate, low accuracy) from genuine uncertainty.

## 6.3 Per-Axis Analysis

Accuracy is stratified by each of the five ambiguity axes. This identifies which types of financial ambiguity are hardest for current models and provides actionable direction for future training.

## 6.4 Longitudinal Comparison

We compare FinInteract results against BrowseComp [10] and FinanceBench [3] trends to verify the interaction stagnation hypothesis: retrieval improves while interaction capability stagnates.

**Table 5: Main results across models and three primary modes (Acc %, $\mathrm{DisE}^{+}_{\mathrm{all}}$, and AxisHit@1 in the interact mode). AHit is undefined when no question is asked. Remaining models in progress.**

| | Answer-only | | | Answer+Search | | | Answer+Search+Interact | | |
|---|---|---|---|---|---|---|---|---|---|
| Model | Acc | $\mathrm{DisE}^{+}$ | AHit | Acc | $\mathrm{DisE}^{+}$ | AHit | Acc | $\mathrm{DisE}^{+}$ | AHit |
| GPT-5 | 0.6 | .012 | – | 6.4 | .110 | – | **20.2** | **.125** | .38 |
| GPT-4o | 0.6 | .009 | – | **12.1** | **.206** | – | 4.6 | .039 | .37 |
| GPT-5-mini | – | – | – | – | – | – | – | – | – |
| Claude-Sonnet-4 | – | – | – | – | – | – | – | – | – |
| Claude-Opus-4 | – | – | – | – | – | – | – | – | – |
| DeepSeek-V3.1 | – | – | – | – | – | – | – | – | – |
| Qwen3-235B | – | – | – | – | – | – | – | – | – |
| GLM-4.5 | – | – | – | – | – | – | – | – | – |
| *Proposed method baseline (GPT-5 backbone)* | | | | | | | | | |
| Axis-Aware ReAct | – | – | – | – | – | – | – | – | – |
| Human (50-inst) | – | – | – | – | – | – | – | – | – |

### 6.5 Mechanistic Case Study: Where Ambiguity Lives in the Network

The behavioral results (§6) show that frontier models default to the naive interpretation rather than clarifying. A natural question is whether the bottleneck is *recognition* (the model never internally registers the ambiguity) or *action* (it registers the ambiguity but answers anyway). Because this requires access to hidden states, we run a small mechanistic probe on two open models as a proxy: **Qwen3-4B-Instruct** (dense, 37 layers) and **Qwen3.5-4B** (hybrid linear/full attention, 33 layers).

*Method.* For each instance we form a within-item contrast: the bare ambiguous question $Q$ versus $Q$ concatenated with its disambiguating context $C$. We define a per-layer *ambiguity direction* as the difference in mean final-token activations between the two conditions, and score any hidden state by its (mean-centered) projection onto that direction. Tracing this score across layers gives a depth profile of how strongly each layer separates an under-specified question from a disambiguated one. An analogous entity–metric direction measures at which depth the model distinguishes the *type* of ambiguity.

*Result.* Figure 1 shows the profiles overlaid for both models (normalized for differing depth and scale). The signal is near-zero through the first ∼60% of layers and rises sharply to a peak in the *final few layers* (layer 35/37 and 32/33 respectively, ≈95–97% relative depth). The entity-scope and metric-definition directions separate with *opposite sign* at the same peak layer in both models. Despite different attention mechanisms, layer counts, and library versions, the two architectures agree: financial-query ambiguity, and its axis, is linearly encoded late in the stack.

*Finding 6: Open models linearly represent both ambiguity and its axis in their final layers, yet still default rather than clarify.* The internal representation exists and is type-specific, replicating across two distinct architectures; the behavioral gap is therefore better read as a failure to *act* on a represented ambiguity than a failure to register it.

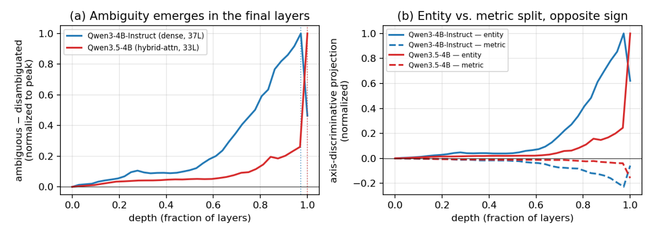


**Figure 1: Cross-model depth profile of the ambiguity signal. (a) Ambiguous–disambiguated separation rises to a peak in the final layers of both models. (b) Entity-scope (solid) and metric-definition (dashed) separate with opposite sign at the top of the stack. Curves are normalized for differing depth (x) and magnitude (y); dotted verticals mark per-model peaks.**

*Scope and caveats.* This is mechanistic evidence on open proxies, not the closed frontier models benchmarked in §6; diff-in-means probing establishes linear decodability, not causal use; and axis-type analysis is statistically supported only for the two well-populated axes (entity_scope, metric_definition). We deliberately do *not* draw behavioral conclusions (e.g. clarification rates) from these open models' generations, as surface cues confound them; the asking-behavior claims come solely from the graded closed-model evaluation in §6.

## 7 ERROR ANALYSIS AND AXIS-AWARE CLARIFICATION

### 7.1 Error Taxonomy

Beyond aggregate accuracy, we characterize model failures using a seven-type error taxonomy (Table 6) derived from FinInteract's evaluation signals. Each type is diagnosable from existing metrics without additional annotation.

**Table 6: Error taxonomy E1–E7 with diagnostic signals and improvement directions.**

| Type | Description | Improvement direction |
|---|---|---|
| E1 | **Ambiguity blindness**: no interaction, wrong answer | Ambiguity pre-check; train on detection |
| E2 | **Wrong-axis clarification**: asked wrong dimension | Axis classifier; financial taxonomy grounding |
| E3 | **Generic clarification**: vague ask, no axis target | Axis-conditioned clarification templates |
| E4 | **Retrieval dominance**: searched but skipped interaction | Force interpretation check after retrieval |
| E5 | **Clarification misuse**: right question, wrong answer | State-gated answering; stronger grounding |
| E6 | **Evidence grounding**: template-oracle correct, agent wrong | Improved source-grounded extraction |
| E7 | **Over-interaction**: correct but low DisE$^+$ | Penalize via DisE$^+$/AxisHit efficiency signal |

## 7.2 Diagnostic Mapping from Evaluation Signals

The existing evaluation metrics map directly onto error types, enabling failure attribution without additional annotation:

- **Low IR + low Acc** → E1 (ambiguity blindness dominant)
- **High IR + low AxisHit@1** → E2/E3 (interacts but misdirected)
- **High AxisHit@1 + low Acc** → E5/E6 (right axis, wrong grounding)
- **High GenericAskRate** → E3 (lacks financial clarification structure)
- **High WrongAxisRate** → E2 (confidently misdirected)
- **Always-ask ≫ Standard ReAct** → E1 dominant; latent capacity exists
- **Template-oracle ≫ Agent-interact** → "what to ask" is the bottleneck
- **Axis-oracle ≫ Agent-interact** → axis recognition is the bottleneck
- **Template-oracle still low** → E6 residual; evidence extraction still fails

## 7.3 Axis-Aware Clarification ReAct

Standard ReAct leaves the agent free to decide whether and what to ask. We propose **Axis-Aware Clarification ReAct**, a three-component structured interaction policy that directly addresses E1–E3.

*Component 1: Ambiguity Pre-Check.* Before any action, the agent explicitly decides which of the five ambiguity axes are underspecified in the question and initializes an interpretation state $\mathcal{S}$ = {entity, period, metric, basis, vintage}, marking each field as *resolved* or *unknown*.

*Component 2: Axis-Conditioned Clarification.* For each *unknown* field, the agent generates a targeted yes/no question using axis-conditioned templates (one per axis) rather than free-form clarification. The agent emits at most one clarification per turn, prioritizing the most information-theoretically valuable unknown field.

*Component 3: Interpretation-State-Gated Answering.* The agent may only emit a final answer once all required fields are resolved (or the user explicitly declines to clarify). This prevents the *retrieval dominance* pattern (E4) where the agent answers immediately after finding a plausible number in a search result.

*Expected outcomes.* Relative to standard Answer+Search+Interact, Axis-Aware ReAct should reduce GenericAskRate (E3), WrongAxisRate (E2), and IR=0 failures (E1), while improving AxisHit@1 and ultimately Accuracy. Template-oracle provides an upper bound: if Axis-Aware ReAct approaches template-oracle performance, it confirms that axis recognition is the dominant remaining bottleneck after the E1–E3 errors are addressed.

# 8 RELATED WORK

*Financial QA Benchmarks.* Table 7 compares FinInteract to major financial QA benchmarks. FinanceBench [3], FinQA [2], DocFinQA [8], BizBench [4], and FinBen [11] all enforce single-gold-answer conventions and evaluate models in a single turn. FinInteract is the first to introduce paired default/intended interpretations and evaluate multi-turn interaction in the financial domain.

*Ambiguous QA..* AmbigQA [7], ASQA [9], CondAmbigQA [6], and CLAM [5] have matured in the general (Wikipedia) domain. We adapt their formalism to finance, replacing human-judged ambiguity labels with programmatically verifiable XBRL-grounded evidence pairs.

*Interactive Search Agents.* InteractComp [1] is the direct methodological ancestor. We extend its general-domain setting to finance with a domain-specific ambiguity taxonomy and bilingual evaluation. BrowseComp [10] tracks longitudinal retrieval trends; we test whether the interaction-capability gap identified there persists in finance.

# 9 CONCLUSION

We introduced FinInteract, the first bilingual benchmark for evaluating financial search agents on genuinely ambiguous queries. The five-axis financial ambiguity taxonomy, paired default/intended interpretation design, and adversarial verifier-based construction pipeline yield 500 high-quality instances that resist trivial disambiguation. Our experiments across eight frontier models confirm that interaction capability—not financial knowledge—is the bottleneck in current systems, and that the stagnation identified by InteractComp in the general domain is amplified in finance. We release FinInteract with full construction code, passage sources, and evaluation scripts to support reproducible follow-on work.

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**Table 7: Comparison of financial QA benchmarks. ✓ = supported; – = not applicable or not evaluated.**

| Benchmark | Scale | Ambiguity | Interaction | Multi-turn | Bilingual | Evidence | Grounding |
|---|---|---|---|---|---|---|---|
| FinanceBench [3] | 150 | – | – | – | – | human | SEC |
| FinQA [2] | 8,281 | – | – | – | – | human | 10-K/Q |
| DocFinQA [8] | 7,437 | – | – | – | – | human | 10-K/Q |
| BizBench [4] | multi-task | – | – | – | – | auto | mixed |
| FinBen [11] | multi-task | partial | – | – | partial | auto | mixed |
| FinInteract (ours) | 500 | ✓ | ✓ | ✓ | EN+ZH | LLM+human | EDGAR+CNINFO |